

Welcome!

The Metro NExT Workshop is a mathematics pedagogy and professional development workshop geared towards graduate students, postdocs, and early-career mathematicians in the NYC metro area and is open to anyone who are interested in learning about new teaching techniques and in making connections with other math educators.

Schedule

9:30-10:00: Registration and Welcome

Metro NExT Organizers.

10:00-11:00: *Getting Started with Undergraduate Mathematics Research...in the Age of AI*

Benjamin Gaines, Iona University

Working with undergraduates on research can be an exceptionally rewarding experience, but many new faculty aren't sure how to get started. It is our hope that this will serve as a primer on the process, with tips on everything from recruiting students to selecting a project to helping students prepare to present their work. Throughout, we will also highlight ways that the availability of generative AI tools has impacted the undergraduate research experience, and the opportunities and challenges this technology provides.

11:00-12:00: Lightning Talks

12:00-1:00: Lunch

1:00-2:00: *Rethinking Undergraduate Research: From Challenging Assumptions to Expanding Possibilities*

Mercedes Franco, Queensborough Community College

Calls to improve undergraduate mathematics education, particularly during the first two years of college, invite us to reconsider how students experience and engage with mathematics. Undergraduate research (UR) offers one promising way to create opportunities for students to explore meaningful questions, make connections across disciplines, and experience mathematics as a dynamic process of discovery. Yet, access to these experiences remains uneven across institutions and student populations.

Drawing on experiences developing and supporting undergraduate research opportunities for students at 2- and 4-year institutions, as well as facilitating professional development for faculty across institutional contexts, this talk invites participants to consider a broader vision of UR as an inclusive pedagogical practice, examine the preparation and institutional support faculty need to make these experiences meaningful and accessible, and explore approaches to assessing their impact on students, faculty, and departmental cultures.

2:00-3:00: Roundtable Discussion

3:00-3:10: Thank you - Metro NExT co-Coordinators

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MATHEMATICAL ASSOCIATION OF AMERICA

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Invited Speakers

Benjamin Gaines is Associate Professor and Chair of Mathematics and Physics at Iona University. He is also Treasurer of the MAA Metro NY Section, and the recipient of Iona's 2022 Brother Arthur A. Loftus Outstanding Student Research Award, and the MAA Metro NY Section's 2024 Distinguished Teaching Award. His primary research interests include combinatorial games and math education, particularly on questions involving writing in mathematics. In addition to teaching courses throughout the math curriculum, from Mathematical Thinking to Abstract Algebra, he greatly enjoys working with students on research questions in games and statistical data analysis. He is very interested in helping learners of all levels overcome the perception that they are "not math people".

Mercedes Franco is a Professor in the Department of Mathematics & Computer Science at Queensborough Community College (CUNY) and Co-Director of the Center for Undergraduate Research in Mathematics (CURM). She has a M.S. and a Ph.D. in Applied Mathematics from Cornell University and a B.S. in Mathematics from Universidad del Valle (Colombia). In 2014-2016, she served as (founding) campus director of the CUNY Research Scholars Program, a university-wide program that funds year-long, laboratory-based STEM research experiences for associate degree students. As an UR practitioner, she has mentored students from 2- and 4-year colleges/universities in summer and academic-year research using the apprenticeship and research-in-the-classroom models.

Lightning Talk Participants

Monica Deni Morales Hernandez (Adelphi University, Mathematics and Computer Science)
Luis A Zambrano (CUNY Guttman Community College, Mathematics)
Suman Ganguli (New York City College of Technology, Mathematics)
Rebecca Conley (Saint Peter's University, Mathematics and Statistics)
Elizabeth Zollinger (St. Joseph's University, Mathematics and Statistics)
Beatriz Levin (CUNY Guttman Community College, Mathematics)

Workshop Organizers

Monica Morales Hernandez, Adelphi University
Andrew Lee, St. Thomas Aquinas College
Liem Nguyen, Brooklyn College

Other Mathematical Events and Professional Development Opportunities

National MAA Project NExT Fellow
Application Deadline: April 15, 2027. projectnext.maa.org

MAA Metro NY Section Meeting: Spring 2027 (date TBD)
Further information can be found on the section's website: http://sections.maa.org/metrony/

The 5th Northeastern Conference on Research in Undergraduate Mathematics Education (NE RUME) Saturday, November 20, 2021 (online) https://sites.google.com/view/pcrg/events/nerume

Joint Mathematics Meetings 2027. January 12-15 2027, Chicago, IL.

Mathfest 2027. August 4-7 2027, New Orlean, LA.

2027 Hudson River Undergraduate Mathematics Conference, TBD at Skidmore College.